

0620 • PAPER 4

TOPIC

11

Organic chemistry

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56

QUESTIONS

721

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Hydrocarbons, fuels, functional groups and polymers

56 questions • 721 marks

11.1 Hydrocarbons, fuels, functional groups and polymers

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11.1 Hydrocarbons, fuels, functional groups and polymers

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11.1 Hydrocarbons, fuels, functional groups and polymers

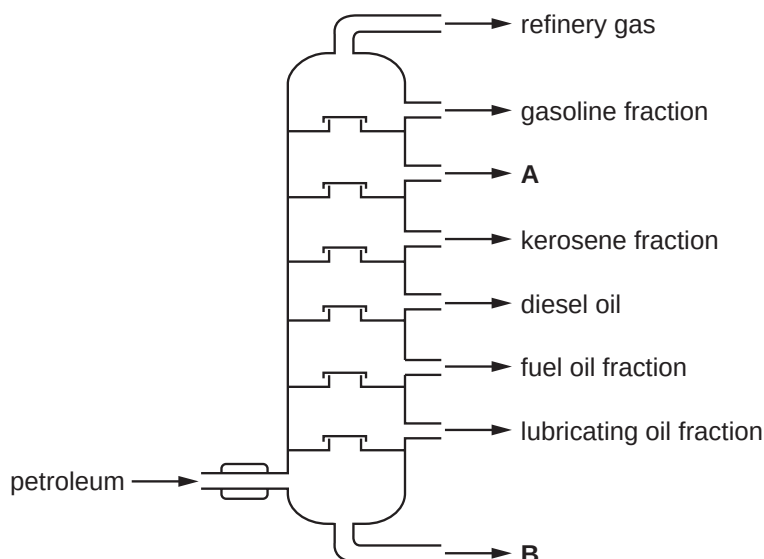
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11.1 Hydrocarbons, fuels, functional groups and polymers

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- 1 Petroleum is a useful natural resource.

The diagram shows how petroleum can be separated into useful substances.



- (a) What is the name of the separation process shown in the diagram?

..... [2]

- (b) Name the fraction leaving at:

A

B [2]

- (c) Refinery gas is a mixture of hydrocarbons.

One refinery gas is butane, C_4H_{10} .

- (i) Suggest the names of **two** other refinery gases.

..... and [2]

- (ii) Write the chemical equation for the complete combustion of butane.

..... [2]

- (iii) Name the toxic gas produced by the incomplete combustion of butane.

..... [1]

(d) Gasoline and kerosene are both fuels. They have different properties.

(i) Describe the differences in the properties given.

viscosity of the fuel

.....

flammability of the fuel

.....

[2]

(ii) What difference in the molecules of gasoline and kerosene causes these differences in properties?

..... [1]

(e) Hydrogen fuel cells can be used to power vehicles.

Write the word equation for the overall reaction that takes place in a hydrogen fuel cell.

..... [1]

[Total: 13]

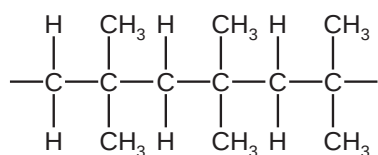
5 There are two types of polymers.

(a) Addition polymers are made from many identical small units.

(i) What is the term used to describe these small units?

..... [1]

(ii) A section of an addition polymer is shown.



Draw the structure of the small unit used to make this addition polymer.

Show all of the atoms and all of the bonds.

[2]

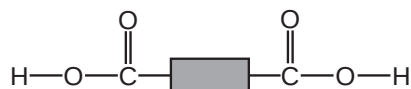
(b) Polyamides are condensation polymers.

What does the term *condensation* mean when used to describe this type of polymer?

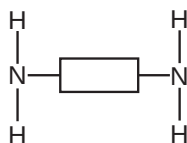
..... [1]

(c) A polyamide can be made from two different molecules.

A simplified structure of octanedioic acid is shown.



A simplified structure of 1,6-diaminohexane is shown.



(i) Complete the diagram to show a section of polyamide manufactured from octanedioic acid and 1,6-diaminohexane. Include all of the atoms and all of the bonds in the linkages.



[3]

(ii) State the name of a synthetic polyamide.

..... [1]

[Total: 8]

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The Periodic Table of Elements

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11 Na sodium 23	12 Mg magnesium 24	13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40	19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131	55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	114 Fl flerovium —	116 Lv livermorium —	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).

5 Ethanol is manufactured by two different processes.

(a) For each process, name the organic reactant and state the type of reaction.

organic reactant type of reaction

organic reactant type of reaction

[4]

(b) Alcohols can be oxidised to form carboxylic acids.

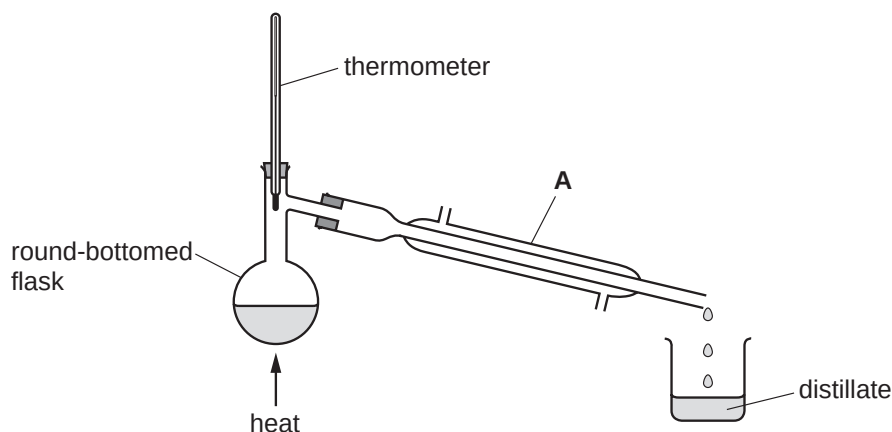
Name a suitable oxidising agent for this reaction.

..... [1]

(c) Alcohols can be partially oxidised to form aldehydes.

Aldehydes are a homologous series of organic compounds.

Partial oxidation is achieved by reacting an alcohol with the oxidising agent in distillation apparatus as shown.



(i) Name apparatus A.

..... [1]

(ii) On the diagram, use **one** arrow to show where water enters apparatus A.

[1]

(d) The table shows some information about aldehydes.

(i) Complete the table.

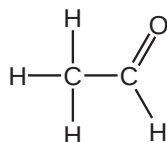
name		ethanal	propanal	butanal
molecular formula	CH ₂ O	C ₂ H ₄ O	C ₃ H ₆ O	

[2]

(ii) Deduce the general formula of aldehydes.

..... [1]

(e) The structural formula of ethanal is shown.

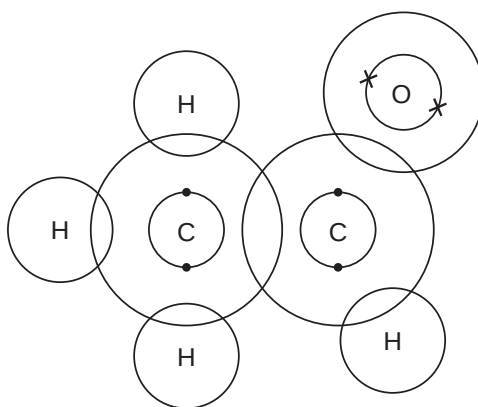


The C=O group in aldehydes is at the end of the carbon chain.
This is a reactive part of the molecule.

(i) What is the name given to the reactive part of any organic molecule?

..... [1]

(ii) Complete the dot-and-cross diagram to show the electron arrangement of a molecule of ethanal. Inner shells have been drawn.



[3]

(f) Propanone belongs to a homologous series called ketones. Ketones have the same C=O group as aldehydes but the C=O group is not at the end of the carbon chain. Propanone has the same molecular formula as propanal, C₃H₆O.

(i) What term is used to describe molecules with different structures but with the same molecular formula?

..... [1]

(ii) Suggest the structure of propanone, C₃H₆O. Show all of the atoms and all of the bonds.

[2]

[Total: 17]

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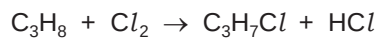
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The Periodic Table of Elements

Group																
I	II						III	IV	V	VI	VII	VIII				
		1 H hydrogen 1											2 He helium 4			
		Key atomic number atomic symbol name relative atomic mass														
3 Li lithium 7	4 Be beryllium 9															
11 Na sodium 23	12 Mg magnesium 24	5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20									
		13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40									
19 K potassium 39	20 Ca calcium 40	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84									
37 Rb rubidium 85	38 Sr strontium 88	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131									
55 Cs caesium 133	56 Ba barium 137	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —									
87 Fr francium —	88 Ra radium —	114 Fl flerovium —	115 Nh nihonium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganesson —	119 Uue unbinilium —									
lanthanoids		57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
actinoids		89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

- 6 (a) Propane reacts with chlorine in a photochemical reaction as shown.



- (i) What type of reaction is this?

..... [1]

- (ii) What condition is needed for this photochemical reaction to occur?

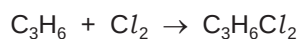
..... [1]

- (iii) Draw **two** structural isomers of compounds with the formula $\text{C}_3\text{H}_7\text{Cl}$.
Show all of the atoms and all of the bonds.



[2]

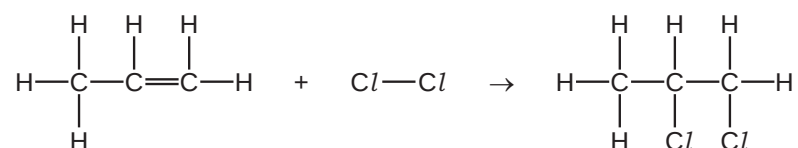
- (b) Propene reacts with chlorine in an addition reaction as shown.



- (i) State why this is an addition reaction.

..... [1]

(ii) The structures of the reactants and products of this reaction are shown.



Some bond energies are shown in the table.

bond	bond energy in kJ/mol
C–C	347
C=C	612
C–H	413
C–Cl	339
Cl–Cl	242

Calculate the energy change for the reaction between propene and chlorine using the following steps.

- Calculate the energy needed to break the bonds.

..... kJ

- Calculate the energy released when bonds are formed.

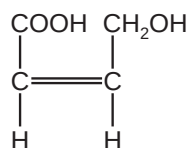
..... kJ

- Calculate the energy change for the reaction between propene and chlorine.

..... kJ/mol
[3]

- (c) There are three functional groups in compound A.

compound A



- (i) Name the homologous series of compounds that contains the following structures.

C=C

–OH

–COOH

[3]

- (ii) What would you observe when compound A is added to:

aqueous bromine

aqueous sodium carbonate?

[2]

- (d) Compound A can be used as a single monomer to produce two different polymers.

- (i) Draw **one** repeat unit of the addition polymer formed from compound A.

[2]

- (ii) What type of condensation polymer is formed from compound A?

..... [1]

[Total: 16]

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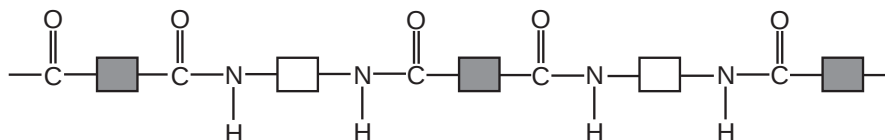
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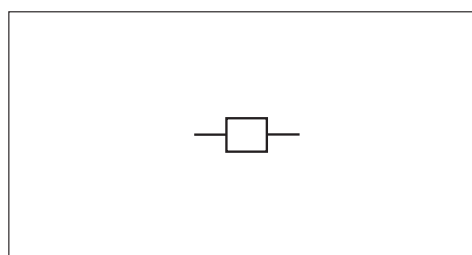
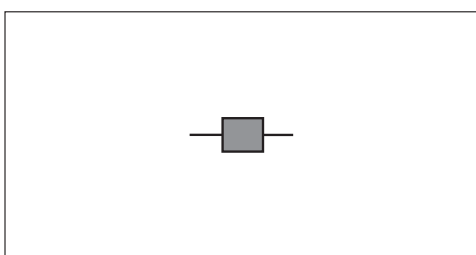
The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

- 8 (a) Part of the synthetic polymer, nylon, is shown in the diagram.



- (i) Circle **one** amide linkage **on the diagram**. [1]

- (ii) Complete the structures of the **two** monomers that react to form nylon.



[2]

- (iii) Name the other product formed when nylon is produced.

..... [1]

- (b) Items made from nylon are often disposed of by burying them in the ground. This is called landfill.

Why is the disposal of nylon using landfill a problem?

.....
..... [1]

- (c) Give the name of a natural polymer.

..... [1]

[Total: 6]

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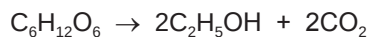
The Periodic Table of Elements

Group																		
I	II	1 H hydrogen 1										III	IV	V	VI	VII	VIII	
		Key atomic number atomic symbol name relative atomic mass															2 He helium 4	
3 Li lithium 7	4 Be beryllium 9											5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20	
11 Na sodium 23	12 Mg magnesium 24											13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40	
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84	
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131	
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids		72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —
87 Fr francium —	88 Ra radium —	89–103 actinoids		104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	114 Fl flerovium —	116 Lv livermorium —	—	—	—	—
lanthanoids															69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175	
actinoids															101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —	

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

- 7 (a) Ethanol can be manufactured by two different methods.

Method 1: fermentation of a sugar, $C_6H_{12}O_6$



Method 2: reaction of ethene with steam



- (i) Give **one** advantage of using fermentation compared with Method 2.

.....
..... [1]

- (ii) Give **one** disadvantage of using fermentation compared with Method 2.

.....
..... [1]

- (b) Ethanol reacts with acidified potassium manganate(VII) to form water and a product that turns litmus red.

- (i) State the name of the product that turns the litmus red.

..... [1]

- (ii) State the type of reaction that ethanol undergoes when it reacts with acidified potassium manganate(VII).

..... [1]

- (c) Ethanol reacts with methanoic acid to form an ester.

- (i) Name the ester formed in this reaction.

..... [1]

- (ii) Draw the structure of the ester formed.
Show all of the atoms and all of the bonds.

[1]

(d) The table shows the melting points of ethanol and sodium chloride.

substance	melting point/°C
ethanol	−114
sodium chloride	801

The difference in melting points is due to differences in attractive forces between particles in these substances.

Name the type of attractive force in each substance, which is responsible for the difference in melting points.

ethanol

sodium chloride

[2]

[Total: 8]

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																	He
																	helium
																	4
3	Li	4	Be														10
	lithium		beryllium														Ne
	7		9														neon
																	20
11	Na	12	Mg														18
	sodium		magnesium														Ar
	23		24														argon
																	40
19	K	20	Ca														36
	potassium		calcium														Kr
	39		40														krypton
																	84
37	Rb	38	Sr														54
	rubidium		strontium														Xe
	85		88														xenon
																	131
55	Cs	56	Ba														86
	caesium		barium														Rn
	133		137														radon
																	–
87	Fr	88	Ra														–
	francium		radium														–
																	–

Key

atomic number
atomic symbol
name
relative atomic mass

lanthanoids

actinoids

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).

4 Alkenes and alkanes are homologous series of compounds containing carbon and hydrogen atoms.

(a) State the name of the type of compound made from carbon and hydrogen atoms only.

..... [1]

(b) Alkenes take part in addition reactions.

(i) Describe what is meant by the term *addition reaction*.

..... [1]

(ii) Draw the structure of the product made in the addition reaction between propene and bromine. Show all of the atoms and all of the bonds.

[2]

(iii) Describe the colour change seen when propene is added to aqueous bromine.

from to [2]

(iv) Draw the structures of molecules of **two** different alkenes which both undergo an addition reaction with steam to form butan-2-ol. Show all of the atoms and all of the bonds.

[2]

(c) Propane undergoes a substitution reaction with chlorine.

Write the chemical equation for the reaction between one molecule of propane and one molecule of chlorine.

..... [2]

[Total: 10]

5 This question is about alcohols, carboxylic acids and esters.

(a) Ethanol will react with hot aqueous potassium manganate(VII) to form ethanoic acid.

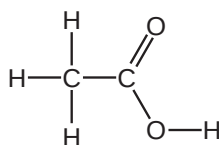
(i) State the other condition needed for this reaction to take place.

..... [1]

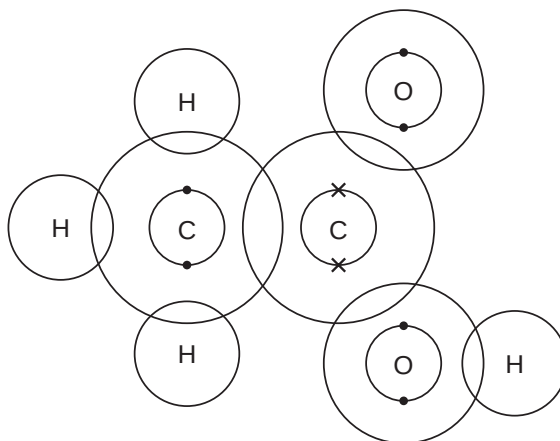
(ii) State the type of chemical change that happens to the ethanol during this reaction.

..... [1]

(iii) The structure of ethanoic acid is shown.



Complete the dot-and-cross diagram to show the electron arrangement in a molecule of ethanoic acid.



[3]

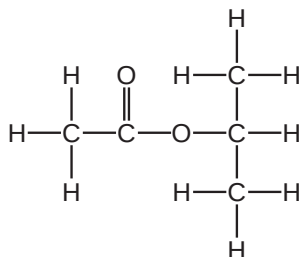
- (b) Ethanoic acid is a weak acid and hydrochloric acid is a strong acid.

Complete the table to show the similarities and differences in the properties of samples of these two acids of equal concentration.

	dilute ethanoic acid	dilute hydrochloric acid
extent of dissociation		
colour after adding universal indicator solution		
observation when magnesium ribbon is added		

[6]

- (c) Ethanoic acid will react with an alcohol to form the ester shown.



- (i) Name the **other** product formed when ethanoic acid reacts with an alcohol to make this ester.
..... [1]
- (ii) Give **one** condition needed when ethanoic acid reacts with the alcohol to make this ester.
..... [1]
- (iii) Draw the structure of the alcohol which was added to ethanoic acid to make this ester. Show all of the atoms and all of the bonds.

[2]

- (d) Polyesters can be manufactured from carboxylic acids and alcohols.

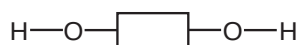
Hexanedioic acid has the structure: $\text{HOOC}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{COOH}$.

This structure can be simplified as shown.



Ethanediol has the structure: $\text{HO}-\text{CH}_2-\text{CH}_2-\text{OH}$.

This structure can be simplified as shown.



The functional groups are found at the end of each molecule.

- (i) State what is meant by the term *functional group*.

..... [1]

- (ii) Determine the empirical formula of hexanedioic acid.

..... [1]

- (iii) Calculate the percentage by mass of oxygen present in ethanediol.

Give your answer to the nearest whole number.

..... % [2]

- (iv) Complete the diagram to show a section of polyester manufactured from hexanedioic acid and ethanediol. Include all of the atoms and all of the bonds in the linkages.



[2]

- (v) State the name of a polyester.

..... [1]

[Total: 22]

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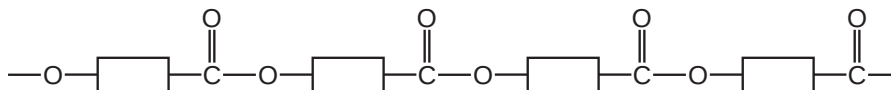
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19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84		
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lanthanoids		57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175			
actinoids		89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —			

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

(b) Part of a polyester chain is shown. This polyester is made from one monomer.



(i) On the diagram draw a ring around one unit of the polymer that is repeated. [1]

(ii) Name the type of polymerisation that produces polyesters.

..... [1]

(iii) Complete the diagram to show the structure of the monomer used to produce this polyester. Show all of the atoms and all of the bonds in the functional groups.



[2]

(c) A polyamide is made from the two monomers shown.



Complete the diagram to show a section of the polyamide made from the two monomers. Show all of the atoms and all of the bonds in the linkages.



[2]

(d) Naturally occurring polyamides are constituents of food.

(i) State the name given to naturally occurring polyamides.

..... [1]

(ii) Name the monomers which form naturally occurring polyamides.

..... [1]

[Total: 14]

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11	12											13	14	15	16	17	18										
Na sodium 23	Mg magnesium 24											Al aluminium 27	Si silicon 28	P phosphorus 31	S sulfur 32	Cl chlorine 35.5	Ar argon 40										
19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36										
K potassium 39	Ca calcium 40	Sc scandium 45	Ti titanium 48	V vanadium 51	Cr chromium 52	Mn manganese 55	Fe iron 56	Co cobalt 59	Ni nickel 59	Cu copper 64	Zn zinc 65	Ga gallium 70	Ge germanium 73	As arsenic 75	Se selenium 79	Br bromine 80	Kr krypton 84										
37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54										
Rb rubidium 85	Sr strontium 88	Y yttrium 89	Zr zirconium 91	Nb niobium 93	Mo molybdenum 96	Tc technetium —	Ru ruthenium 101	Rh rhodium 103	Pd palladium 106	Ag silver 108	Cd cadmium 112	In indium 115	Sn tin 119	Sb antimony 122	Te tellurium 128	I iodine 127	Xe xenon 131										
55	56	57–71 lanthanoids	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86										
Cs caesium 133	Ba barium 137											Tl thallium 204	Pb lead 207	Bi bismuth 209	Po polonium —	At astatine —	Rn radon —										
87	88	89–103 actinoids	104	105	106	107	108	109	110	111	112																
Fr francium —	Ra radium —												Fl flerovium —		Lv livermorium —												

lanthanoids																	
57	58	59	60	61	62	63	64	65	66	67	68	69	70	71			
La lanthanum 139	Ce cerium 140	Pr praseodymium 141	Nd neodymium 144	Pm promethium —	Sm samarium 150	Eu europium 152	Gd gadolinium 157	Tb terbium 159	Dy dysprosium 163	Ho holmium 165	Er erbium 167	Tm thulium 169	Yb ytterbium 173	Lu lutetium 175			
actinoids																	
89	90	91	92	93	94	95	96	97	98	99	100	101	102	103			
Ac actinium —	Th thorium 232	Pa protactinium 231	U uranium 238	Np neptunium —	Pu plutonium —	Am americium —	Cm curium —	Bk berkelium —	Cf californium —	Es einsteinium —	Fm fermium —	Md mendelevium —	No nobelium —	Lr lawrencium —			

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).

5 The table shows the names or structures of organic compounds **P** to **U**.

P	Q	R
$ \begin{array}{ccccc} & \text{H} & & \text{H} & & \text{H} \\ & & & & & \\ \text{H} & - \text{C} & - & \text{C} & - & \text{C} & - \text{H} \\ & & & & & \\ & \text{H} & & \text{H} & & \text{H} \end{array} $	propanoic acid	but-1-ene
S	T	U
propan-1-ol	methyl butanoate	$ \begin{array}{ccccc} & \text{H} & & \text{H} & & \text{H} \\ & & & & & \\ & \text{C} & = & \text{C} & - & \text{C} & - \text{H} \\ & & & & & \\ & \text{H} & & & & \text{H} \end{array} $

(a) Give the letters of the organic compounds, **P** to **U**, that are unsaturated hydrocarbons.

..... [2]

(b) Describe the test for an unsaturated hydrocarbon.

test

observations [2]

(c) But-1-ene is an unbranched molecule.

(i) Name the unbranched isomer of but-1-ene.

..... [1]

(ii) Draw the structure of a branched isomer of but-1-ene. Show all of the atoms and all of the bonds.

[1]

(d) Dodecane is an alkane with 12 carbon atoms. Dodecane can be cracked.

(i) Write the formula of dodecane.

..... [1]

(ii) Give the letters of all the organic compounds, **P** to **U**, that can be formed when dodecane is cracked.

..... [2]

- (e) Name the reagent and suggest the conditions needed to convert organic compound **U** into organic compound **S**.

reagent

conditions [3]

- (f) Organic compound **S** can be converted to organic compound **Q** by reaction with an acidified reagent.

- (i) Name the type of chemical change that happens to organic compound **S**.

..... [1]

- (ii) Name the acidified reagent added to organic compound **S**.

..... [1]

- (g) Organic compound **T** is made by reacting two compounds together.

- (i) Name the homologous series that organic compound **T** belongs to.

..... [1]

- (ii) Name the **two** compounds which react together to make organic compound **T**.

Draw the structures of each compound you have named. Show all of the atoms and all of the bonds.

name

structure

name

structure

[4]

- (iii) Deduce the molecular formula of organic compound **T**.

..... [1]

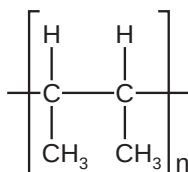
[Total: 20]

6 Polymers are large molecules built up from small molecules.

(a) State the name given to the small molecules from which polymers are made.

..... [1]

(b) The formula of a polymer is shown.



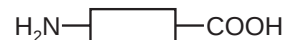
(i) Draw the structure of the small molecule from which this polymer is made. Show all of the atoms and all of the bonds.

[2]

(ii) State the type of polymerisation used to make this polymer.

..... [1]

(c) Three amino acids are shown. They combine to form part of a natural polymer.



(i) Name the type of natural polymer formed when amino acids combine.

..... [1]

(ii) Complete the diagram to show part of the structure of the natural polymer that forms when these three amino acids combine. Show all of the bonds in the linkages.



[3]

(iii) Name the type of chemical reaction that takes place when this natural polymer is converted back to amino acids.

..... [1]

[Total: 9]

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		Key atomic number atomic symbol name relative atomic mass															
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11 Na sodium 23	12 Mg magnesium 24																
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids		72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —
87 Fr francium —	88 Ra radium —	89–103 actinoids		104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	114 Fl flerovium —	116 Lv livermorium —				

lanthanoids	57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
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The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).

6 Molecules **A** and **B** can form condensation polymers.



(a) Each molecule has two identical functional groups.

(i) Name the functional group in **B**.

..... [1]

(ii) Draw the part of the structure of the synthetic polymer that would form when two molecules of **A** and two molecules of **B** combine. Show all of the bonds in the linkages.

[3]

(iii) Name the other product formed when molecules of **A** and **B** undergo polymerisation.

..... [1]

(b) Molecule **A** is a simple sugar unit which can be made by hydrolysis of complex carbohydrates.

(i) Draw part of the complex carbohydrate that could be hydrolysed to make molecules of **A**.

Include **one** linkage and show all of the bonds in the linkage.

[1]

(ii) State **two** sets of conditions which could be used to hydrolyse the complex carbohydrate to form **A**.

1

2

[2]

(iii) Name the technique used to identify the individual sugar units made by the hydrolysis of a complex carbohydrate.

..... [1]

(c) Ethanol can be made from the simple sugar glucose, $C_6H_{12}O_6$.

(i) State the name of this process.

..... [1]

(ii) Complete the chemical equation for this reaction.



[Total: 12]

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																	He
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																	4
3	Li	4	Be														10
	lithium		beryllium														Ne
	7		9														neon
																	20
11	Na	12	Mg														18
	sodium		magnesium														Ar
	23		24														argon
																	40
19	K	20	Ca														36
	potassium		calcium														Kr
	39		40														krypton
37	Rb	38	Sr														84
	rubidium		strontium														Xe
	85		88														xenon
																	131
55	Cs	56	Ba														86
	caesium		barium														Rn
	133		137														radon
																	–
87	Fr	88	Ra														–
	francium		radium														–
																	–

Key

atomic number
atomic symbol
name
relative atomic mass

lanthanoids

actinoids

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).

1 Give the name of the process that is used:

(a) to produce large molecules from monomers

..... [1]

(b) to separate oxygen from liquid air

..... [1]

(c) to make ethanol from glucose

..... [1]

(d) to separate water from aqueous sodium chloride

..... [1]

(e) to produce aluminium from aluminium oxide in molten cryolite

..... [1]

(f) to separate the products of hydrolysis of long chain carbohydrates

..... [1]

(g) to separate an aqueous solution from an undissolved solid.

..... [1]

[Total: 7]

- 5 Alkenes and carboxylic acids are both families of similar compounds with similar chemical properties. Alkenes and carboxylic acids have different reactions.

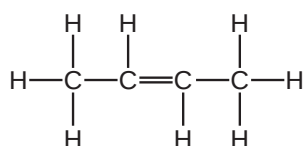
(a) State the term used for a 'family' of similar compounds.

..... [1]

(b) State the general formula of alkenes.

..... [1]

(c) The structure of but-2-ene is shown.



(i) But-2-ene reacts with aqueous bromine in an addition reaction.

Describe the colour change seen when but-2-ene is added to aqueous bromine.

from to [1]

(ii) State what is meant by the term *addition reaction*.

..... [1]

(iii) Write the chemical equation for the reaction between but-2-ene and bromine.

..... [2]

(iv) But-2-ene forms a polymer.

Suggest the name of the polymer formed from but-2-ene.

..... [1]

(v) Name and draw a structural isomer of but-2-ene.

Show all of the atoms and all of the bonds.

name

structure

[2]

(d) Butanoic acid, $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$, is a carboxylic acid.

(i) Deduce the empirical formula of butanoic acid.

..... [1]

(ii) Complete the chemical equation for the reaction of butanoic acid and sodium carbonate, Na_2CO_3 .

$2\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH} + \text{Na}_2\text{CO}_3 \rightarrow \text{.....} + \text{.....} + \text{.....}$ [2]

(iii) Butanoic acid reacts with methanol to form an organic compound and water.

- Name the organic compound formed.

..... [1]

- Draw the structure of the organic compound formed.

Show all of the atoms and all of the bonds.

[2]

[Total: 15]

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37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131	
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids		72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —
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The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).

6 Ethanol, $\text{C}_2\text{H}_5\text{OH}$, belongs to the homologous series called alcohols.

(a) Write the general formula of alcohols.

..... [1]

(b) Explain why ethanol **cannot** be described as a hydrocarbon.

..... [1]

(c) Ethanol can be manufactured from different substances by reaction with steam or by fermentation.

(i) Give the formula of the substance which reacts with steam to form ethanol.

..... [1]

(ii) Name a substance which will undergo fermentation to form ethanol.

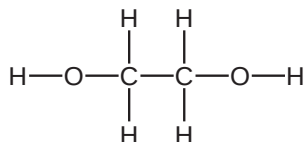
..... [1]

(d) Ethanol is a fuel.

Write the chemical equation for the complete combustion of ethanol.

..... [2]

- (e) Ethane-1,2-diol has two alcohol functional groups.



One molecule of ethane-1,2-diol will react with two molecules of ethanoic acid to form molecule **X**.

X has two ester functional groups and a molecular formula of $\text{C}_6\text{H}_{10}\text{O}_4$.

- (i) State the empirical formula of **X**.

..... [1]

- (ii) Draw the structure of **X**.

Show all of the atoms and all of the bonds.

[2]

- (iii) Name the **other** substance formed in this reaction.

..... [1]

- (f) Each alcohol functional group in ethane-1,2-diol reacts with acidified potassium manganate(VII) to form a different organic compound, **Y**.

- (i) Name the functional groups formed in **Y**.

..... [1]

- (ii) Draw the structure of **Y**.

Show all of the atoms and all of the bonds.

[1]

[Total: 12]

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The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

- 7 (a) Ethanol is a member of the homologous series of alcohols.

Give **two** characteristics of members of a homologous series.

1

2 [2]

- (b) Ethanol can be manufactured from ethene.

Ethene can be made from long chain hydrocarbons such as decane, $C_{10}H_{22}$.

Ethene is then converted into ethanol.

- (i) Name the process used to obtain ethene from long chain hydrocarbons such as decane, $C_{10}H_{22}$.

..... [1]

- (ii) Complete the chemical equation to show the formation of ethene from decane, $C_{10}H_{22}$.



- (iii) Write the chemical equation for the conversion of ethene into ethanol.

..... [1]

- (iv) Name the type of reaction occurring when ethene is converted into ethanol.

..... [1]

- (v) Give **one** condition for the reaction in which ethene is converted into ethanol.

..... [1]

- (c) Ethanol can also be produced by fermentation of carbohydrates such as glucose.

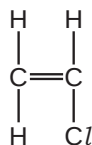
Give **two** advantages of manufacturing ethanol by fermentation compared to manufacturing ethanol from ethene.

1

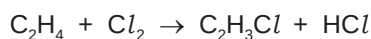
2 [2]

- (d) (i) Under certain conditions ethene can react with chlorine to produce chloroethene.

The structure of chloroethene is shown.



The equation for the chemical reaction is shown.



State the type of chemical reaction between ethene and chlorine that this equation shows.

..... [1]

- (ii) Chloroethene monomers can be converted into a polymer called poly(chloroethene).

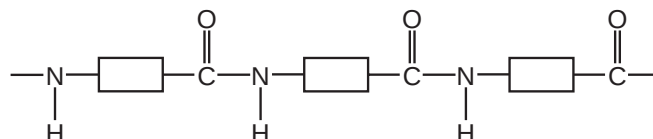
State the type of polymerisation that produces poly(chloroethene) from chloroethene.

..... [1]

- (iii) Draw a section of the poly(chloroethene) molecule made from **two** monomer molecules.

[2]

(e) The structure of part of a polymer is shown.



This polymer is made from one type of monomer only.

Complete the diagram to show the structure of the monomer used to produce this polymer. Show all of the atoms and all of the bonds in the functional groups.



[2]

[Total: 16]

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The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

5 This question is about alkanes and alkenes.

(a) Short-chain alkanes and alkenes can be formed from long-chain alkanes in a chemical reaction.

(i) Name the type of chemical reaction which forms short-chain alkanes and alkenes from long-chain alkanes.

..... [1]

(ii) Decane has 10 carbon atoms. It forms ethane and ethene as the only products in this type of chemical reaction.

Write the chemical equation for this reaction.

..... [3]

(b) Ethane reacts with chlorine at room temperature to form chloroethane, C_2H_5Cl , and one other product.

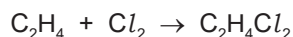
(i) Name the other product formed.

..... [1]

(ii) State the condition needed for this reaction to take place.

..... [1]

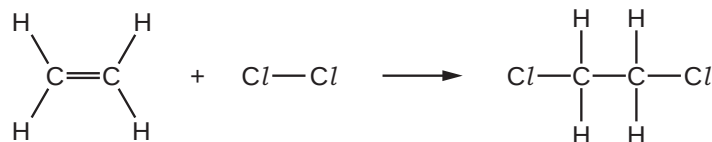
(c) Ethene reacts with chlorine at room temperature to form dichloroethane, $C_2H_4Cl_2$.



(i) State why this is an addition reaction.

..... [1]

(ii) The chemical equation for this reaction can be represented as shown.



The energy change for the reaction is -180 kJ/mol .

Use the bond energies in the table to calculate the bond energy of a $\text{C}-\text{Cl}$ bond, in kJ/mol .

bond	$\text{C}-\text{H}$	$\text{C}=\text{C}$	$\text{Cl}-\text{Cl}$	$\text{C}-\text{C}$
bond energy in kJ/mol	410	610	240	350

Use the following steps.

step 1 Calculate the energy needed to break bonds.

energy needed to break bonds = kJ

step 2 Use your answer in **step 1** and the energy change for the reaction to determine the energy released when bonds are formed.

energy released when bonds form = kJ

step 3 Use your answer in **step 2** and bond energy values to determine the energy of a $\text{C}-\text{Cl}$ bond.

bond energy of a $\text{C}-\text{Cl}$ bond = kJ/mol
[4]

[Total: 11]

6 The names of four esters are listed.

methyl propanoate

ethyl propanoate

propyl propanoate

butyl propanoate

(a) Esters are a family of organic compounds with similar chemical properties. They can be represented by the formula $C_nH_{2n}O_2$.

(i) State the name given to a family of organic compounds with similar chemical properties.

..... [1]

(ii) Explain why members of a family of organic compounds have similar chemical properties.

..... [1]

(iii) State the name given to a formula such as $C_nH_{2n}O_2$.

..... [1]

(iv) Determine the value of 'n' in butyl propanoate.

..... [1]

(b) All four of the esters in the list are liquids at room temperature.

Name the technique used to separate ethyl propanoate from a mixture of the four esters.

..... [2]

(c) All four esters can be made by reacting different alcohols with the same substance.

(i) Name this substance and draw its structure. Show all of the atoms and all of the bonds.

name

structure

[2]

(ii) Name the alcohol used to make methyl propanoate.

..... [1]

(d) Other esters, not in the list, have the same molecular formula as propyl propanoate, but different structures.

(i) State the term used to describe substances with the same molecular formula but different structures.

..... [1]

(ii) Name **two** esters with the same molecular formula as propyl propanoate.

1

2 [2]

(e) Polyesters can be made from the two different molecules shown.



and



(i) Complete the diagram to show a section of the polyester made from these two molecules. Include all of the atoms and all of the bonds in the linkages.



[3]

(ii) Name the type of polymerisation that takes place when this polymer forms.

..... [1]

(iii) Name a polyester.

..... [1]

[Total: 17]

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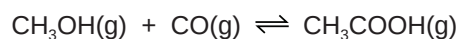
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57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175			
actinoids																	
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —			

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).

- 5 Ethanoic acid is manufactured by the reaction of methanol with carbon monoxide.

An equilibrium mixture is produced.



- (a) State **two** characteristics of an equilibrium.

1

2 [2]

- (b) The purpose of the industrial process is to produce a high yield of ethanoic acid at a high rate of reaction.

The manufacture is carried out at a temperature of 300 °C.

The forward reaction is exothermic.

Use this information to state why the manufacture is **not** carried out at temperatures:

- **below** 300 °C

.....

- **above** 300 °C.

..... [2]

- (c) Complete the table using only the words *increases*, *decreases* or *no change*.

	effect on the rate of the forward reaction	effect on the equilibrium yield of CH ₃ COOH(g)
adding a catalyst		no change
decreasing the pressure		

[3]

- (d) Suggest which of the following metals is a suitable catalyst for the reaction. Give a reason for your answer.

aluminium calcium cobalt magnesium potassium

suitable catalyst

reason

[2]

- (e) Ethanoic acid is a member of the homologous series of carboxylic acids.

State the general formula of this homologous series.

..... [1]

- (f) Draw the structure of the carboxylic acid containing three carbon atoms. Show all of the atoms and all of the bonds.

[2]

- (g) When carboxylic acids react with alcohols, esters are produced.

The formula of ester X is $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOCH}_3$.

- (i) Name ester X.

..... [1]

- (ii) Give the name of the carboxylic acid and the alcohol that react together to produce ester X.

carboxylic acid

alcohol

[2]

- (h) Ester Y has the following composition by mass:

C, 48.65%; H, 8.11%; O, 43.24%.

Calculate the empirical formula of ester Y.

empirical formula = [3]

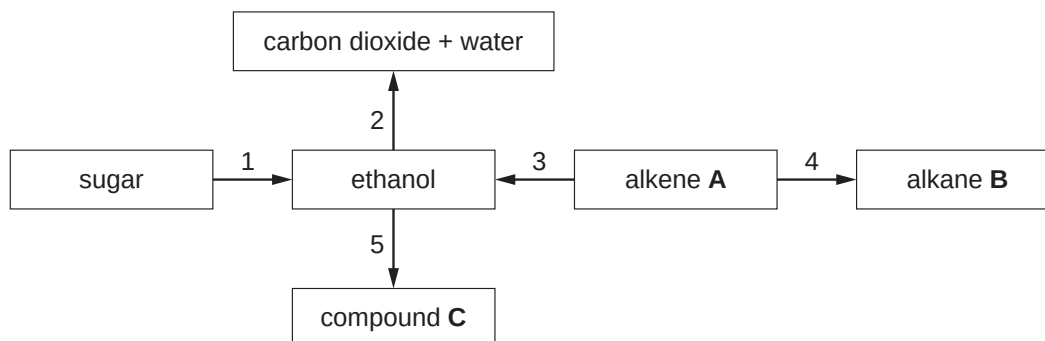
- (i) Ester **Z** has the empirical formula $\text{C}_2\text{H}_4\text{O}$ and a relative molecular mass of 88.

Determine the molecular formula of ester **Z**.

molecular formula = [1]

[Total: 19]

5 The reaction scheme shows five organic reactions, numbered 1 to 5.



(a) Name reaction 1.

..... [1]

(b) Name reaction 2 and write the chemical equation for this reaction.

name

equation [3]

(c) Reaction 3 forms ethanol from alkene A.

(i) Identify alkene A.

..... [1]

(ii) State the type of reaction that occurs during reaction 3.

..... [1]

(iii) State the reagents and conditions needed for reaction 3.

..... [2]

(d) Alkene A is converted into alkane B in reaction 4.

(i) State the reagent and conditions for reaction 4.

..... [3]

(ii) State the general formula of alkanes.

..... [1]

(e) Ethanol is oxidised in reaction 5 by heating it with dilute sulfuric acid and one other reagent.

(i) Identify the other reagent in reaction 5.

..... [1]

(ii) Name the homologous series compound **C** belongs to.

..... [1]

(iii) Draw the structure of compound **C**.

Show all of the atoms and all of the bonds.

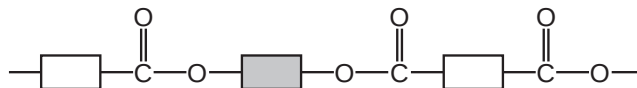
[1]

[Total: 15]

6 This question is about polymers.

(a) Polymer X is a condensation polymer.

Part of the structure of polymer X is shown.



(i) How many molecules of water are produced when this part of polymer X is formed from its monomers?

..... [1]

(ii) Complete the structures of the **two** monomers used to make polymer X.

Show all of the atoms and all of the bonds in the functional groups.



and



[2]

(iii) What type of condensation polymer is X?

..... [1]

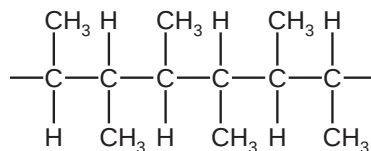
(b) Part of polymer Y has the structure shown.



State the number of different types of monomer needed to make polymer Y.

..... [1]

(c) Part of polymer **Z** has the structure shown.



(i) Draw and name the structure of the monomer which forms polymer **Z**.

Show all of the atoms and all of the bonds.

name

[3]

(ii) Name the chemical process used to make the monomer that forms polymer **Z**.

..... [1]

[Total: 9]

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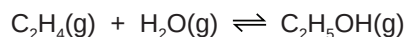
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The Periodic Table of Elements

Group																											
I	II	1 H hydrogen 1					III	IV	V	VI	VII	VIII															
3 Li lithium 7	4 Be beryllium 9	Key atomic number atomic symbol name relative atomic mass					5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20															
													11 Na sodium 23	12 Mg magnesium 24	13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40							
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84										
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131										
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —										
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Lv livermorium —	116 Ts tennessine —	117 Ug unseptennium —	118 Og oganesson —										
lanthanoids													67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175										
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The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).

- 4 Ethanol is made industrially by the reaction of ethene with steam. The reaction occurs at a temperature of 300 °C and a pressure of 60 atmospheres.



A catalyst is used in this reaction.

The forward reaction is exothermic.

- (a) State what is meant by the term *catalyst*.

.....
..... [2]

- (b) Complete the table using only the words *increases*, *decreases* or *no change*.

	effect on the rate of the forward reaction	effect on the equilibrium yield of C ₂ H ₅ OH(g)
increasing the temperature		
decreasing the pressure		

[4]

- (c) Ethanol is a member of the alcohol homologous series.

Members of the same homologous series have the same general formula.

- (i) State the general formula of alcohols.

..... [1]

- (ii) State two **general** characteristics, other than the same general formula, of all homologous series.

1

2 [2]

- (iii) One alcohol containing three carbon atoms is propan-1-ol.

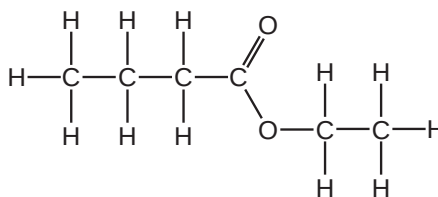
Draw the structure of one **other** alcohol containing three carbon atoms. Show all of the atoms and all of the bonds.

Name the alcohol you have drawn.

name [2]

- (d) When alcohols react with carboxylic acids, esters are produced.

- (i) The structure of ester **X** is shown.



Name ester **X**.

..... [1]

- (ii) Give the name of the alcohol and the carboxylic acid that react together to produce ester **X**.

alcohol

carboxylic acid [2]

- (e) Ester **Y** has the following composition by mass:

C, 58.82%; H, 9.80%; O, 31.37%.

Calculate the empirical formula of ester **Y**.

empirical formula = [3]

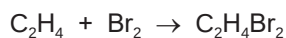
- (f) Ester **Z** has the empirical formula C_3H_6O and a relative molecular mass of 116.

Calculate the molecular formula of ester **Z**.

molecular formula = [1]

[Total: 18]

- 5 Ethene is an alkene which reacts with bromine as shown in the equation.



- (a) Write the general formula of alkenes.

..... [1]

- (b) Describe the colour change seen when ethene is bubbled through aqueous bromine.

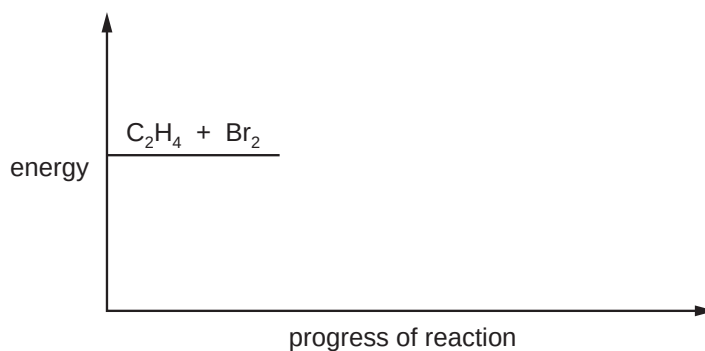
from to [1]

- (c) In this reaction only one product is formed from two reactants.

Name this type of organic reaction.

..... [1]

- (d) Part of the energy profile diagram of this reaction is shown.



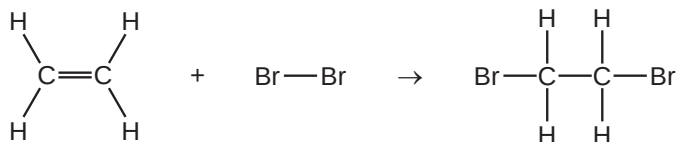
- (i) The reaction is exothermic.

Complete the energy profile diagram for this reaction.
Include:

- the position of the products
- an arrow to show the activation energy, labelled as A
- an arrow to show the energy change for the reaction.

[3]

(ii) The chemical equation for the reaction can be represented as shown.



Some bond energies are given.

bond	bond energy / kJ mol ⁻¹
C-H	410
C=C	610
Br-Br	190
C-C	350
C-Br	290

Use the bond energies in the table to calculate the energy change in this reaction.

Use the following steps.

- Calculate the energy needed to break bonds.

energy = kJ

- Calculate the energy released in making bonds.

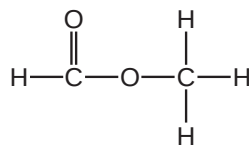
energy = kJ

- Determine the energy change in this reaction.

energy change in this reaction = kJ/mol
[3]

[Total: 9]

6 Ester Y has the structure shown.



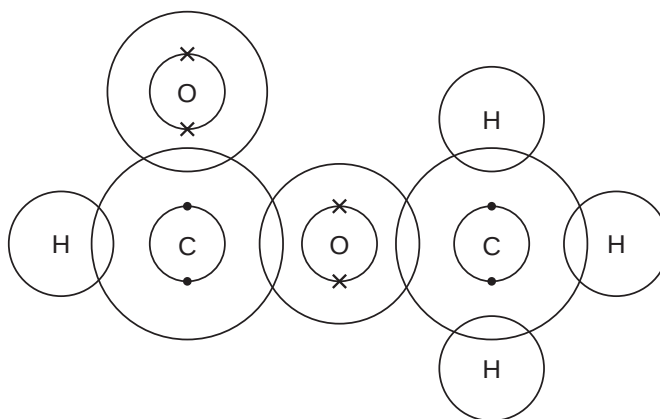
(a) (i) Name ester Y.

..... [1]

(ii) Deduce the empirical formula of ester Y.

..... [1]

(b) Complete the dot-and-cross diagram to show the arrangement of electrons in a molecule of ester Y.



[3]

(c) Ester Y can be made by reacting two organic compounds together.

Name the compounds and draw their structures.

Show all of the atoms and all of the bonds.

name

structure

name

structure

[4]

(d) (i) Describe what is meant by the term *structural isomer*.

.....
..... [2]

(ii) Name a carboxylic acid which is a structural isomer of ester Y.

..... [1]

[Total: 12]

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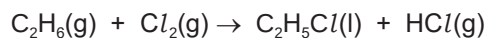
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19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	114 Fl flerovium —	115 Cn copernicium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganesson —
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The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

- 5 Ethane is an alkane which undergoes a photochemical reaction with chlorine as shown.



- (a) Write the general formula of alkanes.

..... [1]

- (b) State why this reaction is described as a photochemical reaction.

..... [1]

- (c) In this reaction, an atom of hydrogen is replaced with a chlorine atom.

State the name of this type of organic reaction.

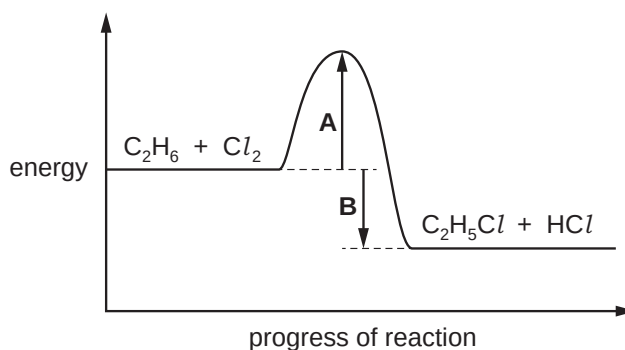
..... [1]

- (d) In this reaction, one of the products is chloroethane.

Name the other product.

..... [1]

- (e) The energy profile diagram of this reaction is shown.



- (i) Name the energy change labelled **A**.

..... [1]

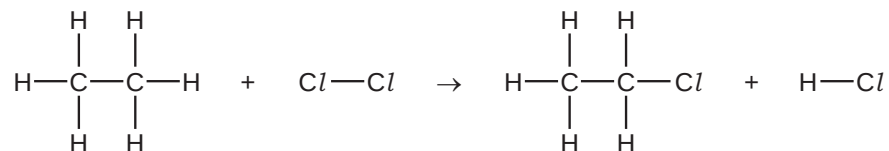
- (ii) Name the energy change labelled **B**.

..... [1]

- (iii) State how the energy profile diagram shows this is an exothermic reaction.

..... [1]

(f) The equation for the reaction can be represented as shown.



Some bond energies are given.

bond	bond energy /kJmol
C-H	410
C-C	350
Cl-Cl	240
C-Cl	340
H-Cl	430

Use the bond energies in the table to calculate the energy change in this reaction.

Use the following steps.

- Calculate the energy needed to break bonds.

energy = kJ

- Calculate the energy released in making bonds.

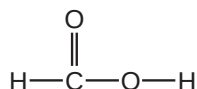
energy = kJ

- Determine the energy change in this reaction.

energy change in this reaction = kJ/mol
[3]

[Total: 10]

- 6 A carboxylic acid **Y** has the structure shown.



- (a) State the general formula of carboxylic acids.

..... [1]

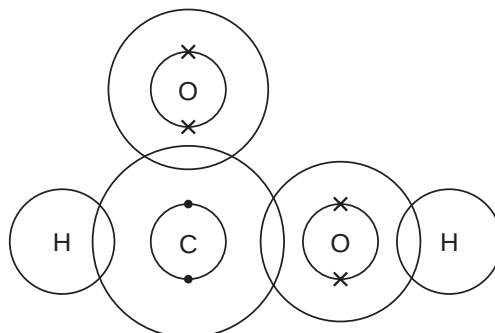
- (b) Name carboxylic acid **Y**.

..... [1]

- (c) Write the molecular formula of carboxylic acid **Y**.

..... [1]

- (d) Complete the dot-and-cross diagram to show the arrangement of electrons in a molecule of carboxylic acid **Y**.



[3]

(e) Carboxylic acid **Y** will react with propan-1-ol, $\text{C}_3\text{H}_7\text{OH}$, to form ester **Z** and one other product.

(i) Name and draw the structure of ester **Z**.

Show all of the atoms and all of the bonds.

name

structure

[3]

(ii) Name the other product formed when carboxylic acid **Y** reacts with propan-1-ol.

..... [1]

(iii) Name:

- an ester which is a structural isomer of ester **Z**

.....

- a carboxylic acid which is a structural isomer of ester **Z**.

.....

[2]

[Total: 12]

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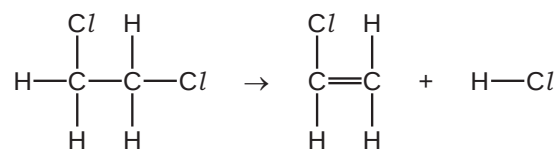
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- 6 (a) Chloroethene ($\text{CH}_2=\text{CHCl}$) can be manufactured from 1,2-dichloroethane ($\text{CH}_2\text{ClCH}_2\text{Cl}$).

The equation can be represented as shown.



- (i) Some bond energies are given.

bond	bond energy in kJ/mol
C—C	350
C=C	610
C—Cl	340
C—H	410
H—Cl	430

Use the bond energies in the table to calculate the energy change, in kJ/mol, of the reaction.

Use the following steps.

- Calculate the energy needed to break bonds.

energy = kJ

- Calculate the energy released when bonds form.

energy = kJ

- Calculate the energy change of the reaction.

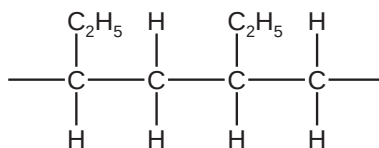
energy change of the reaction = kJ/mol
[3]

- (ii) Deduce whether the energy change for this reaction is exothermic or endothermic.

Give a reason for your answer.

.....
..... [1]

- (b) Part of a synthetic polymer is shown.



- (i) State the number of monomer units that are needed to make the part of the polymer shown.

..... [1]

- (ii) Name and draw the structure of the monomer used to make this polymer. Show all of the atoms and all of the bonds.

name

structure

[3]

- (iii) State the empirical formula of the polymer.

..... [1]

(c) Proteins are natural polymers.

Proteins are broken down into amino acids. The process is similar to how complex carbohydrates are broken down to give simple sugars.

(i) Name the type of reaction in which proteins are broken down into amino acids.

..... [1]

(ii) Name **two** types of substance that are used to break down proteins into amino acids.

1

2 [2]

(iii) Amino acids are colourless.

A sample containing a mixture of amino acids is separated. Each amino acid is detected and identified.

- Name the process used to separate the amino acids.

.....

- Name the type of substance used to detect the amino acids.

.....

- Give the symbol of the value used to determine the identity of each amino acid after separation and detection.

..... [3]

(d) Proteins are natural polymers. Proteins contain amide linkages.

Synthetic polyamides also contain amide linkages.

(i) Name a synthetic polyamide.

..... [1]

(ii) Identify the **two** functional groups present in the monomers used to produce synthetic polyamides.

1

2 [2]

[Total: 18]

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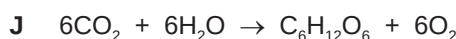
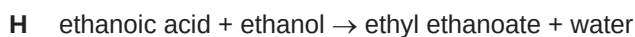
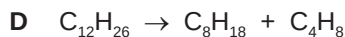
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11 Na sodium 23	12 Mg magnesium 24											13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40	
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84	
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131	
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids		72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —
87 Fr francium —	88 Ra radium —	89–103 actinoids		104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	114 Fl flerovium —		116 Lv livermorium —	—		
lanthanoids		57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175		
actinoids		89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —		

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

1 Some symbol equations and word equations, **A** to **J**, are shown.



Use the equations to answer the questions that follow.

Each equation may be used once, more than once, or not at all.

Give the letter, **A** to **J**, for the equation that represents:

(a) a neutralisation reaction [1]

(b) a precipitation reaction [1]

(c) the formation of an ester [1]

(d) photosynthesis [1]

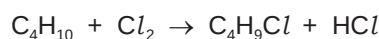
(e) fermentation [1]

(f) cracking. [1]

[Total: 6]

7 This question is about organic compounds.

(a) Butane reacts with chlorine in a photochemical reaction.



(i) State the meaning of the term photochemical.

..... [1]

(ii) An organic compound with the formula $\text{C}_4\text{H}_9\text{Cl}$ is formed when one molecule of butane reacts with one molecule of chlorine.

Draw the displayed formulae of **two** possible structural isomers with the formula $\text{C}_4\text{H}_9\text{Cl}$ formed in this reaction.

[2]

(b) The structure of compound **A** is shown in Fig. 7.1.

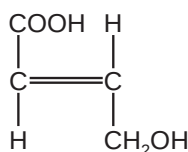


Fig. 7.1

(i) Deduce the molecular formula of compound **A**.

..... [1]

(ii) There are three functional groups in compound **A**.

Name the homologous series of compounds that contain the following functional groups:

—C=C—

—OH

—COOH

[3]

(iii) State what is observed when compound **A** is added to:

aqueous bromine

aqueous sodium carbonate.

[2]

(iv) Compound **A** can be used as a single monomer to produce two different polymers.

Draw **one** repeat unit of the addition polymer formed from compound **A**.

[2]

(v) Compound **A** can be converted into a dicarboxylic acid.

Name the type of condensation polymer formed from a dicarboxylic acid and a diol.

..... [1]

[Total: 12]

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The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

6 Carboxylic acids can be converted to esters.

(a) Name the ester formed when butanoic acid, $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$, reacts with ethanol, $\text{CH}_3\text{CH}_2\text{OH}$.

..... [1]

(b) Identify the other product formed in this reaction.

..... [1]

(c) Deduce the empirical formula of the ester formed.

..... [1]

(d) PET is a polyester. Part of the structure of PET is shown in Fig. 6.1.

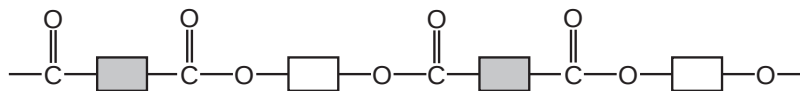


Fig. 6.1

(i) Circle **one** repeat unit of this polymer. [1]

(ii) Draw the structures of the monomers which make up PET. Draw the functional groups using displayed formulae.

[2]

(iii) State the type of polymerisation used in making PET.

..... [1]

[Total: 7]

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The Periodic Table of Elements

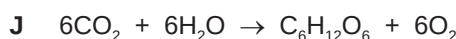
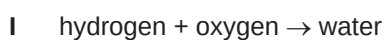
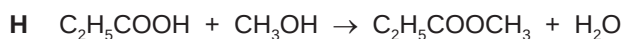
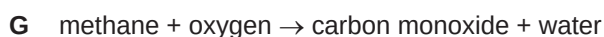
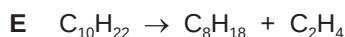
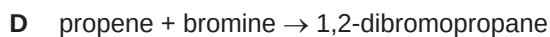
Group																	
I	II	Key					III	IV	V	VI	VII	VIII					
		atomic number atomic symbol relative atomic mass					1 H hydrogen 1						2 He helium 4				
3 Li lithium 7	4 Be beryllium 9						5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20					
11 Na sodium 23	12 Mg magnesium 24						13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40					
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131
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lanthanoids

actinoids

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

1 Some symbol equations and word equations, **A** to **J**, are shown.



Use the equations to answer the questions that follow.

Each equation may be used once, more than once, or not at all.

Give the letter, **A** to **J**, for the equation which represents:

(a) photosynthesis [1]

(b) an addition reaction [1]

(c) a precipitation reaction [1]

(d) incomplete combustion [1]

(e) a displacement reaction [1]

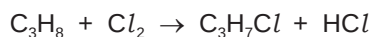
(f) a substitution reaction. [1]

[Total: 6]

Question 2 starts on the next page.

7 This question is about organic compounds.

(a) Propane and chlorine react at room temperature. An equation for the reaction is shown.



(i) State the condition required for this reaction.

..... [1]

(ii) Draw the displayed formulae of **two** structural isomers with the formula $\text{C}_3\text{H}_7\text{Cl}$.

[2]

(b) Alkenes are a homologous series of hydrocarbons.

(i) State **two** characteristics that all members of the same homologous series have in common.

1

2 [2]

(ii) Addition polymers are made from alkenes.

Complete Fig. 7.1 to show one repeat unit of the addition polymer formed from but-2-ene.

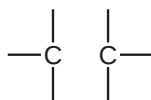


Fig. 7.1

[2]

- (c) A repeat unit of a condensation polymer is shown in Fig. 7.2.
The polymer is made from two monomers.

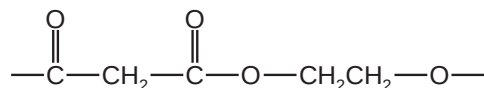
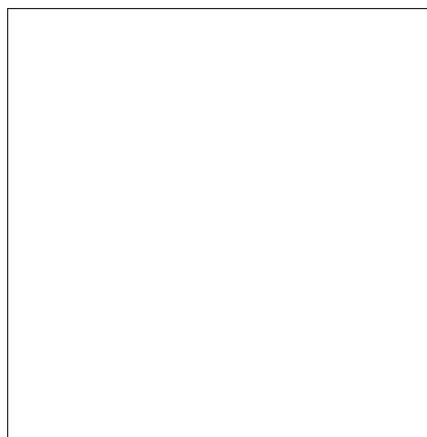
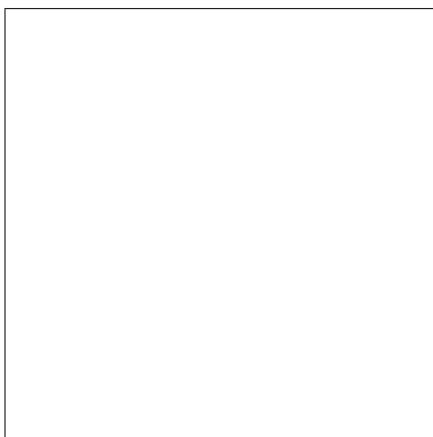


Fig. 7.2

- (i) Draw the structures of the monomers used to produce the polymer in Fig. 7.2.



[2]

- (ii) Name the **type** of condensation polymer in Fig. 7.2.

..... [1]

- (iii) Name the **two** homologous series to which the monomers in (i) belong.

1

2

[2]

[Total: 12]

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Group																		
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actinoids												99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —		

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

6 Ethanol is manufactured by **two** methods:

method 1 fermentation of aqueous glucose

method 2 catalytic addition of steam to an alkene.

(a) Method 1 takes place at room temperature and pressure.

State **two** other conditions needed in method 1.

1

2 [2]

(b) (i) State the typical temperature and pressure used in method 2.

temperature °C

pressure kPa [2]

(ii) Name the alkene used in method 2.

..... [1]

(iii) State why the reaction in method 2 is referred to as an addition reaction.

..... [1]

(c) The catalyst in method 2 is phosphoric acid, H_3PO_4 . Dilute phosphoric acid is a weak acid which contains phosphate ions, PO_4^{3-} .

(i) State what is meant by the term acid.

..... [1]

(ii) State the meaning of weak in the term weak acid.

..... [1]

(iii) Determine the oxidation number of phosphorus in the PO_4^{3-} ion.

Show your working.

oxidation number = [2]

(d) Give **one** advantage of each method of production of ethanol.

method 1

method 2 [2]

(e) Ethanol can be converted to ethanoic acid by reacting it with an acidified oxidising agent.

(i) Name the acidified oxidising agent.

..... [1]

(ii) State, in terms of redox, what type of reagent ethanol is in this reaction.

..... [1]

(f) Ethanoic acid reacts with calcium to form a salt and one other product.

(i) Name the salt.

..... [1]

(ii) Write the formula of the salt.

..... [1]

(iii) Identify the other product.

..... [1]

[Total: 17]

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The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

- 6 A homologous series is a family of organic compounds whose members have similar chemical properties.

(a) Give **two** characteristics that are the **same** for all members of a homologous series.

1

2 [2]

(b) In terms of structure, state how one member of a homologous series differs from the next member of that homologous series.

..... [1]

(c) **A**, **B** and **C** are organic compounds.

A has the molecular formula $C_{12}H_{24}$.

B has the name tetradecane.

C has three carbon atoms and is in the homologous series with the general formula $C_nH_{2n+1}COOH$.

(i) Name the homologous series each organic compound belongs to.

A

B

C [3]

(ii) Name **C** and draw its displayed formula.

name

displayed formula

[2]

- (d) Amino acids are a homologous series where each member has the general structure shown in Fig. 6.1.

The R side chain contains carbon and hydrogen atoms only.

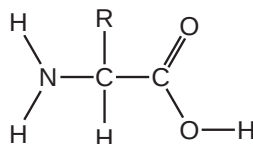


Fig. 6.1

- (i) An amino acid has a relative molecular mass of 103.

Deduce the formula of the R side chain in this amino acid.

Show your working.

..... [2]

- (ii) State the name given to the natural polyamides formed from amino acid monomers.

..... [1]

[Total: 11]

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actinoids		89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —			

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

- 6 (a) Esters are members of a homologous series of organic compounds.

Give **two** characteristics that are the **same** for all members of a homologous series.

1

2 [2]

- (b) Ester X has the structure shown in Fig. 6.1.

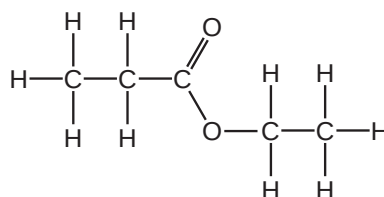


Fig. 6.1

Name ester X.

..... [1]

- (c) (i) Ester Y has the structural formula $\text{HCOOCH}_2\text{CH}_2\text{CH}_3$.

Name the alcohol and the carboxylic acid used to make ester Y.

alcohol

carboxylic acid [2]

- (ii) State the molecular formula of ester Y.

..... [1]

- (d) Ester Z has the molecular formula $\text{C}_4\text{H}_8\text{O}_2$.

State the empirical formula of ester Z.

..... [1]

(e) Polymers containing ester linkages are known as polyesters.

Polyamides are another type of polymer. Nylon is a polyamide.

The structure of nylon is shown in Fig. 6.2.

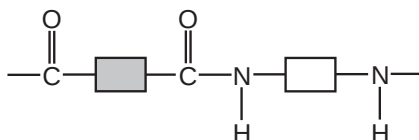


Fig. 6.2

(i) State the term used to describe the type of polymerisation used to produce polyesters and polyamides.

..... [1]

(ii) Complete Fig. 6.3 to show the structures of the monomers used to produce nylon. Show all of the atoms and all of the bonds.

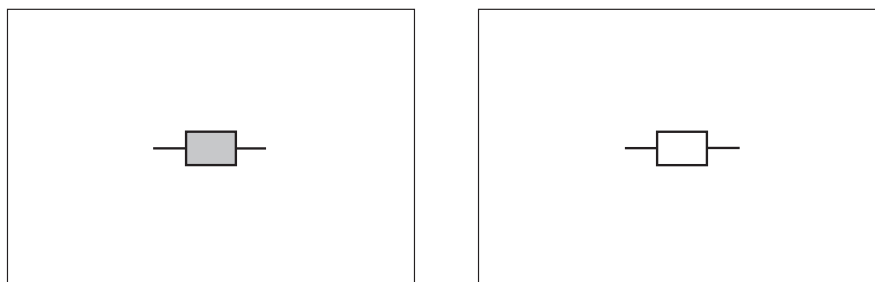


Fig. 6.3

[2]

(f) Naturally occurring polyamides are found in food.

(i) State the name given to naturally occurring polyamides.

..... [1]

(ii) Name the type of monomer which forms naturally occurring polyamides.

..... [1]

[Total: 12]

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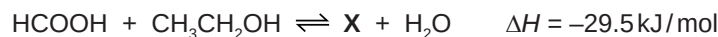
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Group																																									
I	II	1 H hydrogen 1										III	IV	V	VI	VII	VIII																								
3 Li lithium 7		Key atomic number atomic symbol name relative atomic mass										5 B boron 11					6 C carbon 12					7 N nitrogen 14					8 O oxygen 16					9 F fluorine 19					10 Ne neon 20				
												13 Al aluminum 27					14 Si silicon 28					15 P phosphorus 31					16 S sulfur 32					17 Cl chlorine 35.5					18 Ar argon 40				
19 K potassium 39		20 Ca calcium 40		21 Sc scandium 45		22 Ti titanium 48		23 V vanadium 51		24 Cr chromium 52		25 Mn manganese 55		26 Fe iron 56		27 Co cobalt 59		28 Ni nickel 59		29 Cu copper 64		30 Zn zinc 65		31 Ga germanium 73		32 Ge germanium 73		33 As arsenic 75		34 Se selenium 79		35 Br bromine 80		36 Kr krypton 84							
37 Rb rubidium 85		38 Sr strontium 88		39 Y yttrium 89		40 Zr zirconium 91		41 Nb niobium 93		42 Mo molybdenum 96		43 Tc technetium —		44 Ru ruthenium 101		45 Rh rhodium 103		46 Pd palladium 106		47 Ag silver 108		48 Cd cadmium 112		49 In indium 115		50 Sn tin 119		51 Sb antimony 122		52 Te tellurium 128		53 I iodine 127		54 Xe xenon 131							
55 Cs caesium 133		56 Ba barium 137		57–71 lanthanoids		72 Hf hafnium 178		73 Ta tantalum 181		74 W tungsten 184		75 Re rhenium 186		76 Os osmium 190		77 Ir iridium 192		78 Pt platinum 195		79 Au gold 197		80 Hg mercury 201		81 Tl thallium 204		82 Pb lead 207		83 Bi bismuth 209		84 Po polonium —		85 At astatine —		86 Rn radon —							
87 Fr francium —		88 Ra radium —		89–103 actinoids		104 Rf rutherfordium —		105 Db dubnium —		106 Sg seaborgium —		107 Bh bohrium —		108 Hs hassium —		109 Mt meitnerium —		110 Ds darmstadtium —		111 Rg roentgenium —		112 Cn copernicium —		113 Nh nihonium —		114 Fl flerovium —		115 Mc moscovium —		116 Lv livermorium —		117 Ts tennessine —		118 Og oganeson —							
lanthanoids																67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175																					
actinoids																89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —											

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).

- 4 The equation for the reaction between methanoic acid and ethanol in the presence of a catalyst can be represented as shown.



X represents the ester formed.

- (a) (i) In the equation, methanoic acid is represented by the formula HCOOH.

Name this type of formula.

..... [1]

- (ii) Write the empirical formula of methanoic acid.

..... [1]

- (b) Name and draw the displayed formula of ester X.

name

displayed formula

[3]

- (c) The reaction is reversible and reaches an equilibrium within a closed system.

- (i) State what is meant by the term closed system.

..... [1]

- (ii) State **two** characteristics of an equilibrium.

1

.....

2

.....

[2]

- (iii) Complete Table 4.1 to show the effect, if any, on the concentration of **X** at equilibrium for each change of condition.

Table 4.1

change of condition	effect on the concentration of X at equilibrium
temperature is decreased	
concentration of HCOOH is decreased	
concentrations of both HCOOH and CH ₃ CH ₂ OH are decreased	
the catalyst is removed	

[4]

[Total: 12]

5 Butane and but-1-ene are colourless gases at room temperature and pressure.

(a) Suggest why but-1-ene diffuses quicker than butane.

..... [1]

(b) Identify the products formed when butane undergoes complete combustion.

..... [1]

(c) One molecule of butane reacts with one molecule of chlorine in the presence of ultraviolet light. During the reaction, one hydrogen atom in butane is replaced by one chlorine atom.

(i) Name the type of reaction which needs ultraviolet light.

..... [1]

(ii) State the purpose of ultraviolet light during this reaction.

..... [1]

(iii) Name the type of reaction which takes place when one atom of chlorine replaces one atom of hydrogen.

..... [1]

(iv) Determine how many different structural isomers can form during this reaction.

..... [1]

(d) When but-1-ene reacts with steam, **two** possible products form.

(i) Identify the type of catalyst which is used in this reaction.

..... [1]

(ii) Name and draw the displayed formulae of the **two** possible products.

product 1	product 2
name	name
displayed formula	displayed formula

[4]

(e) But-1-ene undergoes polymerisation.

(i) State the type of polymerisation but-1-ene undergoes.

..... [1]

(ii) Draw part of the polymer molecule to show **three** repeat units.

[3]

[Total: 15]

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The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

1 Name the process that is used to:

(a) convert sulfur dioxide into sulfur trioxide in the manufacture of sulfuric acid

..... [1]

(b) obtain water from aqueous sodium chloride

..... [1]

(c) extract aluminium from purified bauxite

..... [1]

(d) separate petroleum into useful substances

..... [1]

(e) produce ethanol from aqueous glucose

..... [1]

(f) manufacture alkenes and hydrogen from large alkane molecules

..... [1]

(g) separate a mixture of soluble coloured substances.

..... [1]

[Total: 7]

7 Many organic compounds contain carbon and hydrogen only.

(a) (i) Organic compound **A** has the following composition by mass.

C, 82.76%; H, 17.24%

Calculate the empirical formula of compound **A**.

empirical formula = [3]

(ii) Compound **B** has the empirical formula CH_2 and a relative molecular mass of 70.

Determine the molecular formula of compound **B**.

molecular formula = [1]

(b) Fig. 7.1 shows a section of polymer **Q**.

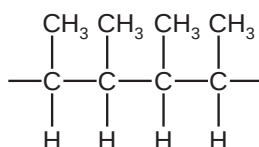


Fig. 7.1

- Draw the displayed formula of the monomer that forms polymer **Q**.

- Name the monomer used to form polymer **Q**.

.....
[3]

- (c) Propene, C_3H_6 , can be produced by heating $C_{11}H_{24}$. The products of the reaction are propene, hydrogen and one other product in a 1 : 1 : 1 mole ratio.

Complete the symbol equation for this reaction.



- (d) Carboxylic acids and esters contain carbon, hydrogen and oxygen only.

An ester **X** and a carboxylic acid **Y** both contain 3 carbon atoms.

X and **Y** have the same molecular formula.

- (i) State the name given to compounds with the same molecular formula but different structural formulae.

..... [1]

- (ii) Esters are made by the reaction between carboxylic acids and alcohols.

Ester **X** is methyl ethanoate.

Name the carboxylic acid and the alcohol used to make methyl ethanoate.

carboxylic acid

alcohol [2]

- (iii) Draw the displayed formula of carboxylic acid **Y**. Name the carboxylic acid.

name [2]

[Total: 13]

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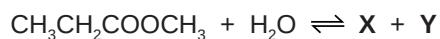
Group																			
I	II	Key										III	IV	V	VI	VII	VIII		
		atomic number atomic symbol name relative atomic mass										1 H hydrogen 1							2 He helium 4
3 Li lithium 7	4 Be beryllium 9											5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20		
11 Na sodium 23	12 Mg magnesium 24											13 Al aluminum 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40		
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84		
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xeron 131		
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids		72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —		
87 Fr francium —	88 Ra radium —	89–103 actinoids		104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og ognesson —		
lanthanoids																			
57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175					
actinoids																			
89 Ac actinium	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —					

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

- 4 A carboxylic acid reacts with an alcohol to produce an ester and water.

Under certain conditions, this reaction can be reversed so an ester reacts with water to produce a carboxylic acid **X** and an alcohol **Y**.

The reaction reaches an equilibrium.



The forward reaction is endothermic.

- (a) Deduce the empirical formula of the ester.

..... [1]

- (b) Name the ester.

..... [1]

- (c) Name carboxylic acid **X** and draw its displayed formula.

name

displayed formula

[2]

- (d) Name alcohol **Y** and give its structural formula.

name

structural formula

[2]

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(e) Complete Table 4.1 to show the effect, if any, for each change of condition.

Table 4.1

change of condition	effect on the concentration of carboxylic acid X at equilibrium
temperature is decreased	
concentration of $\text{CH}_3\text{CH}_2\text{COOCH}_3$ is decreased	
more alcohol Y is added	
a catalyst is added	

[4]

(f) At the beginning of the reaction between the ester and water, no carboxylic acid is present in the reaction mixture.

(i) Suggest how the pH of the reaction mixture changes from the start of the reaction until equilibrium is reached.
Assume alcohols and esters are neutral.

pH at start of reaction

pH at equilibrium

[2]

(ii) Identify the ion that causes the change in pH.

..... [1]

(iii) Name an indicator which can be used to follow the change in pH.

..... [1]

[Total: 14]

6 Glucose is involved in two processes.

(a) Glucose, $C_6H_{12}O_6$, is made in plants from carbon dioxide and water.

(i) Name this process.

..... [1]

(ii) Write the symbol equation for this process.

..... [1]

(iii) State **two** essential conditions needed for this process to happen.

1

2 [2]

(b) Glucose is converted to ethanol.

(i) Name this process.

..... [1]

(ii) Name the **other** product formed when glucose is converted to ethanol.

..... [1]

(c) Ethanol is made by reacting ethene with steam in an industrial process.

(i) State the conditions and type of catalyst used in this industrial production of ethanol.

temperature and units

pressure and units

type of catalyst used [3]

(ii) Explain why this reaction is an addition reaction.

..... [1]

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(iii) Complete the dot-and-cross diagram in Fig. 6.1 of a molecule of ethanol.

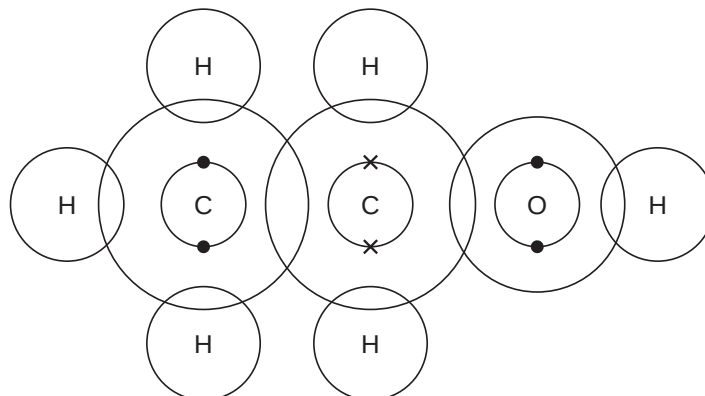


Fig. 6.1

[3]

[Total: 13]



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The Periodic Table of Elements

Group																						
I	II	1 H hydrogen 1										III	IV	V	VI	VII	VIII					
		Key atomic number atomic symbol name relative atomic mass																				
3 Li lithium 7	4 Be beryllium 9																5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20
11 Na sodium 23	12 Mg magnesium 24																13 Al aluminum 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84					
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131					
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids		72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —				
87 Fr francium —	88 Ra radium —	89–103 actinoids		104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganeson —				
lanthanoids		57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175						
actinoids		89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —						

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).

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7 Many organic compounds contain carbon and hydrogen only.

(a) (i) An organic compound **A** has the following composition by mass.

C, 83.33%; H, 16.67%

Calculate the empirical formula of compound **A**.

empirical formula = [3]

(ii) Compound **B** has the empirical formula C_2H_5 and a relative molecular mass of 58.

Determine the molecular formula of compound **B**.

molecular formula = [2]

(b) Fig. 7.1 shows a section of a polymer formed from an alkene.

(i) Identify the functional group in alkenes that reacts when alkenes form polymers.

..... [1]

(ii) A section of a polymer is shown in Fig. 7.1.

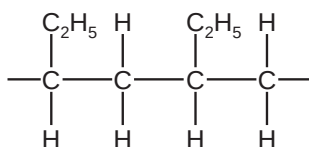


Fig. 7.1

- Draw the displayed formula of the monomer that forms this polymer.

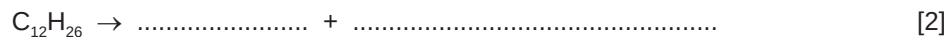
- Name the monomer used to form this polymer.

..... [3]

(c) Alkenes are produced by cracking alkanes.

When $C_{12}H_{26}$ is cracked, the products are ethene and an alkane which form in a 2 : 1 mole ratio.

Write a symbol equation for this reaction.



(d) (i) State the general formula for alcohols.

..... [1]

(ii) Draw the displayed formula of **one** alcohol with the molecular formula C_3H_8O . Name the alcohol you have drawn.

name of alcohol [2]

[Total: 14]

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5 Hydrogen is the first element of the Periodic Table.

(a) Hydrogen is used in fuel cells to produce electricity in vehicles.

(i) Name the substance which combines with hydrogen in a fuel cell.

..... [1]

(ii) Give **one** advantage and **one** disadvantage of using fuel cells instead of gasoline in vehicle engines.

advantage

disadvantage [2]

(b) Hydrogen gas can be made from petroleum by a two-step procedure.

step 1 Petroleum is separated into different components.

step 2 Large molecules obtained in **step 1** are converted into smaller molecules including hydrogen gas.

(i) Name the process used in **step 1**.

..... [1]

(ii) Name the process used in **step 2**.

..... [1]

(c) Organic compounds contain hydrogen atoms.

Calculate the number of hydrogen atoms in 44.0g of the ester methyl propanoate, $\text{CH}_3\text{CH}_2\text{COOCH}_3$.

One mole of $\text{CH}_3\text{CH}_2\text{COOCH}_3$ contains 6.02×10^{23} molecules.

Give your answer in standard form.

number of hydrogen atoms = [4]



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(d) For each of the homologous series shown, name a member that contains **six** hydrogen atoms.

- alkanes
- alkenes
- alcohols
- carboxylic acids

[4]

(e) Unsaturated alkenes are converted into saturated alkanes by reaction with hydrogen gas.

(i) State why alkenes and alkanes are hydrocarbons.

..... [1]

(ii) State why alkenes are unsaturated.

..... [1]

(iii) Name the catalyst needed to convert alkenes into alkanes.

..... [1]

(iv) Explain why the conversion of alkenes into alkanes is an addition reaction.

..... [1]

[Total: 17]



- 1 The formulae of the molecules **A** to **I** are shown in Table 1.1.

Table 1.1

molecule	formula
A	C_2H_4
B	$\text{C}_2\text{H}_5\text{OH}$
C	CO
D	CO_2
E	Cl_2
F	NO_2
G	N_2
H	O_2
I	SO_2

Answer the following questions about the molecules, **A** to **I**.
Each letter may be used once, more than once or not at all.

State which of the molecules **A** to **I**:

- (a) is an element with a triple bond [1]
- (b) is a product of photosynthesis [1]
- (c) is used as a fuel [1]
- (d) turns limewater milky [1]
- (e) undergoes a substitution reaction with alkanes [1]
- (f) is a colourless liquid at r.t.p. [1]
- (g) is unsaturated [1]
- (h) is 21% of clean, dry air [1]
- (i) is a reactant in the Haber process. [1]

[Total: 9]



5 Alkenes are manufactured by cracking larger alkane molecules.

(a) State the source of the large alkane molecules used in cracking.

..... [1]

(b) State **two** conditions needed for cracking large alkane molecules.

1

2 [2]

(c) When one molecule of dodecane, $C_{12}H_{26}$, is cracked, three molecules of but-1-ene and one other product are formed.

(i) Use molecular formulae to complete the symbol equation for this reaction.

$C_{12}H_{26} \rightarrow \dots + \dots$ [2]

(ii) Suggest the type of chemical reaction which happens during cracking.

..... [1]

(d) Propene will undergo polymerisation.

(i) Suggest the name of the polymer formed from propene.

..... [1]

(ii) Draw part of this polymer molecule to show **three** repeat units.

[3]

(iii) State the type of polymerisation propene undergoes.

..... [1]

[Total: 11]



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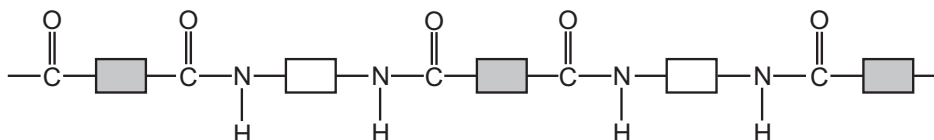
Question 6 starts on the next page.



6 Polyamides and polyesters are polymers.

Polyamides can occur naturally or can be manufactured.

(a) Part of the structure of a polyamide is shown in Fig. 6.1.

**Fig. 6.1**

- (i) On Fig. 6.1, draw a circle around **one** amide linkage. [1]
- (ii) Complete Fig. 6.2 to show the structures of the **two** monomers needed to make the polymer in Fig. 6.1. Show all of the atoms and all of the bonds in the functional groups.

**Fig. 6.2**

- (iii) Name the other product formed in this polymerisation. [2]
- [1]
- (iv) State the term given to natural polyamides. [1]
- [1]
- (v) Name the type of monomers which are used to make natural polyamides. [1]
- [1]





(vi) One of the monomers which forms part of a natural polyamide has **three** carbon atoms.

Complete Fig. 6.3 to show the displayed formula of this monomer.

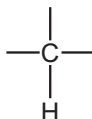


Fig. 6.3

[3]

(b) PET is a polyester.

(i) Name the **two** types of monomer molecules needed to make polyesters.

..... and [2]

(ii) Draw part of the structure of PET which shows **two** repeat units.

Show all of the atoms and all of the bonds in the linkages.

[3]

[Total: 14]

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The Periodic Table of Elements

Group																			
I	II	Key										III	IV	V	VI	VII	VIII		
		atomic number atomic symbol name relative atomic mass										1 H hydrogen 1						2 He helium 4	
3 Li lithium 7	4 Be beryllium 9											5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20		
11 Na sodium 23	12 Mg magnesium 24											13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40		
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84		
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131		
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids		72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —	
87 Fr francium —	88 Ra radium —	89–103 actinoids		104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganeson —	
lanthanoids		57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175			
actinoids		89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —			

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).

6 This question is about hydrocarbons.

(a) State the meaning of the term hydrocarbon.

..... [1]

(b) Propene, C_3H_6 , can be made from long-chain alkanes such as dodecane.
Dodecane contains 12 carbon atoms.

(i) Deduce the molecular formula of dodecane.

..... [1]

(ii) Name the type of reaction that occurs when long-chain alkanes are converted into shorter chain alkenes.

..... [1]

(c) Propene is an unsaturated hydrocarbon.
Propene reacts with bromine.

(i) State the meaning of the term unsaturated.

..... [1]

(ii) Write the molecular formula of the product formed when propene reacts with bromine.

..... [1]

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(d) A styrene molecule is represented as shown in Fig. 6.1.

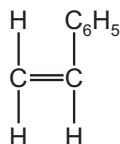


Fig. 6.1

- (i) The molecular formula of styrene is C_8H_8 .

Determine the empirical formula of styrene.

..... [1]

- (ii) Styrene can be polymerised into poly(styrene).

State the type of polymerisation that occurs when styrene is converted into poly(styrene).

..... [1]

- (iii) Draw the structure of **one** repeat unit of poly(styrene). Include all of the atoms and all of the bonds.

The C_6H_5 group should be represented as C_6H_5 .

[2]

[Total: 9]





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The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).

- 6 The structural formulae of two compounds, **A** and **B**, are shown.

A	B
$\text{CH}_2=\text{CHCH}_3$	$\text{CH}_3\text{CH}=\text{CHCH}_3$

A and **B** are members of the same homologous series.

- (a) Give **two** reasons why the structural formulae of **A** and **B** show they are members of the same homologous series.

1

2 [2]

- (b) Explain why **A** and **B** are both hydrocarbons.

..... [1]

- (c) Write the symbol equation for the complete combustion of **A**.

..... [2]

- (d) Deduce the empirical formula of **A**.

..... [1]

- (e) Name compound **B**.

..... [1]

- (f) A structural isomer of **B** is a member of the same homologous series.

Draw the displayed formula of this structural isomer of **B**.

[1]



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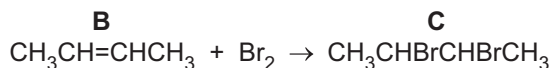
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- (g) Compound **B** reacts with aqueous bromine at room temperature to form product **C**.
The equation is shown.



- (i) State why this is an addition reaction.

..... [1]

- (ii) Describe the colour change in aqueous bromine during this reaction.

from to [1]

- (iii) Name product **C**.

..... [1]

- (h) Under certain conditions, **one** mole of **B** reacts with oxygen to form **two** moles of carboxylic acid **D**.

Carboxylic acid **D** has **two** carbon atoms.

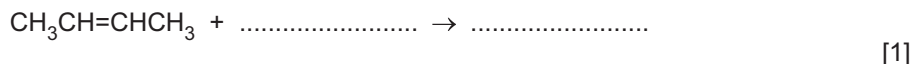
- (i) Draw the displayed formula of carboxylic acid **D**.

[1]

- (ii) Name carboxylic acid **D**.

..... [1]

- (iii) Complete the symbol equation for this reaction.



[Total: 14]

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The Periodic Table of Elements

[illegible]

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

lanthanoids

actinoids

- 1 A list of substances is shown.

bauxite	calcium oxide	ethanol	graphite	methane
nitrogen	oxygen	propane	propene	sulfur dioxide

Answer the questions using the list of substances.

Each substance may be used once, more than once, or not at all.

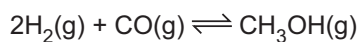
State which of the substances:

- (a) is a basic oxide
..... [1]
- (b) contains carbon atoms only
..... [1]
- (c) is manufactured by fermentation
..... [1]
- (d) is produced by the decomposition of vegetation
..... [1]
- (e) contains aluminium oxide
..... [1]
- (f) causes acid rain
..... [1]
- (g) is a simple molecule with 11 atoms
..... [1]
- (h) is produced when limestone thermally decomposes in the blast furnace
..... [1]
- (i) is a gas that is approximately 21% of clean, dry air
..... [1]
- (j) is a monomer in addition polymerisation.
..... [1]

[Total: 10]



- 5 Methanol, CH₃OH, is manufactured by the reaction between hydrogen and carbon monoxide. An equilibrium mixture is produced.



- (a) State what happens to the concentration of CH₃OH when the reaction is at equilibrium.

Explain your answer in terms of rate of forward and reverse reactions.

.....

 [2]

- (b) The reaction is carried out at a pressure of 75 atmospheres and a temperature of 250 °C.

- (i) State **two** disadvantages of using a pressure **below** 75 atmospheres.

1
 2 [2]

- (ii) Complete Table 5.1 using **only** the words **increases**, **decreases** or **no change**.

Table 5.1

	effect on the concentration of CH ₃ OH(g) at equilibrium	effect on the rate of the reverse reaction
catalyst is added		

[2]

- (iii) If a temperature of more than 250 °C is used, the yield of methanol decreases.

State what can be deduced about the forward reaction.

..... [1]

- (iv) Suggest which of the elements from the list is a suitable catalyst for the reaction. Give a reason for your answer.

barium carbon copper potassium sulfur

catalyst

reason

[2]



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(c) Methanol is a member of the homologous series of alcohols.

(i) State **two** characteristics of all members of a homologous series.

1

2 [2]

(ii) State the molecular formula of an alcohol that contains five carbon atoms.

..... [1]

(d) Carboxylic acids react with alcohols to form esters.

(i) Draw the displayed formula of an ester which contains three carbon atoms.

[2]

(ii) Butyl ethanoate is an ester.

Name the alcohol and the carboxylic acid that react to produce butyl ethanoate.

alcohol

carboxylic acid [2]

(e) An organic compound has the following composition by mass:

C, 64.87%; H, 13.51%; O, 21.62%.

Calculate the empirical formula of the compound.

empirical formula = [2]

[Total: 18]

[Turn over]



- 6 Ethene, C_2H_4 , is the first member of a family of similar compounds which contains the alkene functional group.

(a) State the term for a family of similar compounds which contain the same functional group.

..... [1]

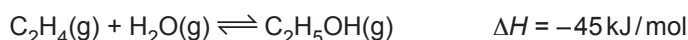
(b) Determine the difference in relative molecular mass between C_2H_4 and the next member in this family of similar compounds.

..... [1]

(c) Write the symbol equation for the complete combustion of C_2H_4 .

..... [2]

(d) C_2H_4 reacts with steam to form ethanol.



The process happens in a closed system and the reaction reaches an equilibrium.
The conditions for this process are 300°C and 60 atm pressure. H_3PO_4 is used as a catalyst.

- (i) Complete Table 6.1 to show the effect, if any, on the concentration of $C_2H_4(g)$ when the following changes to the conditions are applied.

Only use the words **increases**, **decreases** or **no change**.

Table 6.1

change to condition	effect on the concentration of $C_2H_4(g)$ at equilibrium
temperature is decreased	
some $C_2H_5OH(g)$ is removed	
pressure is increased	
a more effective catalyst is used	

[4]

- (ii) Explain, in terms of collision theory, why the rate of the forward reaction increases if the temperature increases.

.....

 [3]

- (e) Compound **B** has the displayed formula shown in Fig. 6.1.

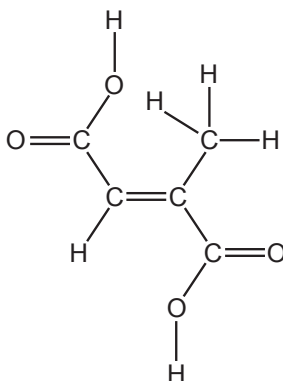


Fig. 6.1

- (i) Deduce the molecular formula of compound **B**.
..... [1]
- (ii) State why compound **B** is unsaturated.
..... [1]
- (iii) Draw the structure of **one** repeat unit of the polymer formed when compound **B** undergoes addition polymerisation.
..... [2]
- (iv) Explain why 1 mol of compound **B** reacts with 2 mol of sodium hydroxide, NaOH.
.....
..... [1]
- (v) Calculate the volume, in cm³, of 0.250 mol/dm³ NaOH that reacts with 0.100 mol of compound **B**.

volume = cm³ [2]

[Total: 18]

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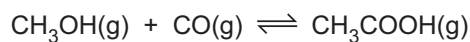
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actinoids																		

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).

- 5 Ethanoic acid is manufactured in the reaction between methanol and carbon monoxide.

An equilibrium mixture is produced.



- (a) State **two** characteristics of an equilibrium.

1

2 [2]

- (b) Both a high yield of ethanoic acid and a high rate of reaction are needed.

The reaction is carried out at 300 °C.

The forward reaction is exothermic.

- (i) State the disadvantage of using a temperature:

- below 300 °C

.....

- above 300 °C.

..... [2]

- (ii) Complete Table 5.1 using **only** the words **increases**, **decreases** or **no change**.

Table 5.1

	effect on the equilibrium concentration of CH ₃ COOH(g)	effect on the rate of the forward reaction
catalyst is added		increases
pressure is increased		

[3]

- (iii) Suggest which element from the list is a suitable catalyst for the reaction.

Give a reason for your answer.

aluminium carbon cobalt magnesium sodium

catalyst

reason [2]



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(c) Ethanoic acid is a member of the carboxylic acid homologous series.

(i) Name the carboxylic acid that contains only one carbon atom.

..... [1]

(ii) State the molecular formula of a carboxylic acid that contains four carbon atoms.

..... [1]

(d) Carboxylic acids react with alcohols to form esters.

(i) Draw the displayed formula of the ester which contains two carbon atoms.

[2]

(ii) Propyl butanoate is an ester.

Name the carboxylic acid and alcohol that react to produce propyl butanoate.

carboxylic acid

alcohol

[2]

(e) An organic compound has the following composition by mass:

C, 58.82%; H, 9.80%; O, 31.38%.

Calculate the empirical formula of the compound.

empirical formula = [3]

[Total: 18]



5 This question is about the homologous series of alcohols.

(a) A homologous series is a family of organic compounds whose members have the same general formula.

(i) State the general formula for alcohols.

..... [1]

(ii) Give **one other** characteristic that is the same for all members of a homologous series.

..... [1]

(b) Ethanol can be manufactured by two methods:

- **method 1** uses glucose as the starting material
- **method 2** uses ethene as the starting material.

(i) Complete Table 5.1.

Table 5.1

	method 1 glucose as starting material	method 2 ethene as starting material
typical temperature used / °C		
two other essential conditions	1	1
	2	2

[6]

(ii) Write the symbol equation for the reaction in **method 1**.

..... [2]

(iii) Write the symbol equation for the reaction in **method 2**.

..... [2]

(c) Butane-1,4-diol has the structural formula $\text{HO}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{OH}$.

(i) Deduce the molecular formula of butane-1,4-diol.

..... [1]

(ii) Butane-1,4-diol reacts with ethanoic acid.

Determine the number of moles of ethanoic acid which react fully with **one** mole of butane-1,4-diol.

..... [1]



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(d) Butanedioic acid has the structural formula $\text{HOOC}-\text{CH}_2-\text{CH}_2-\text{COOH}$.

(i) Deduce the empirical formula of butanedioic acid.

..... [1]

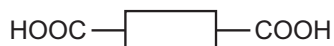
(ii) Name the gas formed when butanedioic acid reacts with sodium.

..... [1]

(e) Butane-1,4-diol can be represented as shown.



Butanedioic acid can be represented as shown.



Butane-1,4-diol reacts with butanedioic acid to form a polymer.

(i) Draw **two** repeat units of the polymer formed from the reaction of butane-1,4-diol with butanedioic acid.

Show all the atoms and all the bonds in the ester linkages.

[3]

(ii) State the type of polymerisation when butane-1,4-diol reacts with butanedioic acid.

..... [1]

[Total: 20]





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		atomic number atomic symbol name relative atomic mass										1 H hydrogen 1							
3 Li lithium 7	4 Be beryllium 9											5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20		
11 Na sodium 23	12 Mg magnesium 24											13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40		
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84		
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131		
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —		
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganeson —		

lanthanoids	57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
actinoids	89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



5 This question is about the homologous series of alkanes.

Three general characteristics of alkanes are:

- they are generally unreactive because they do **not** have a functional group
- they show trends in their physical properties, such as melting points and boiling points
- they have similar chemical properties.

(a) Describe **two other** general characteristics of the homologous series of alkanes.

1.....
2..... [2]

(b) (i) State the trend in the boiling points of the alkanes as the carbon chain length increases.

..... [1]

(ii) Name **one other** physical property of alkanes that shows a trend, other than melting points and boiling points.

Describe this trend as the carbon chain length of alkanes increases.

name
description [2]

(c) State why alkanes are described as saturated.

..... [1]



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(d) Propane has the structural formula $\text{CH}_3\text{CH}_2\text{CH}_3$.

When propane undergoes a monosubstitution reaction with chlorine at room temperature, **two** organic products can be formed.

(i) State **one** condition needed for this substitution reaction to take place.

..... [1]

(ii) Give the structural formula and state the name of each organic product formed.

- structural formula of product 1

name of product 1

- structural formula of product 2

name of product 2

[4]

(iii) The two products formed are structural isomers of each other.

Define the term structural isomers.

.....

..... [1]

[Total: 12]



- 6 Compound **E** shown in Fig. 6.1 is an ester.

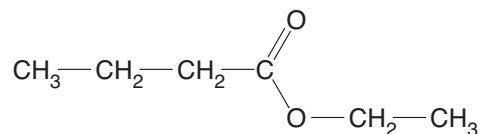


Fig. 6.1

- (a) On Fig. 6.1, draw a circle around all the atoms which make up the ester linkage. [1]

- (b) Name compound **E**.

..... [1]

- (c) Deduce the empirical formula of compound **E**.

..... [1]

- (d) Compound **E** is produced when two different compounds react together in the presence of a catalyst.

- (i) Name the **two** compounds that react together to produce compound **E**.

1

2 [2]

- (ii) Suggest a suitable catalyst for this reaction.

..... [1]

- (e) Determine the number of unbranched esters which have the molecular formula $\text{C}_5\text{H}_{10}\text{O}_2$.

..... [1]

- (f) All unbranched esters with the molecular formula $\text{C}_5\text{H}_{10}\text{O}_2$ combust completely to form carbon dioxide and water only.

Write the symbol equation for the complete combustion of $\text{C}_5\text{H}_{10}\text{O}_2$.

..... [1]

[Total: 8]





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The Periodic Table of Elements

Group											
I	II										
		1 H hydrogen 1									
		Key									
		atomic number atomic symbol name relative atomic mass									
3 Li lithium 7	4 Be beryllium 9										
11 Na sodium 23	12 Mg magnesium 24										
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201
87 Fr francium	88 Ra radium	89–103 actinoids	104 Rf rutherfordium	105 Db dubnium	106 Sg seaborgium	107 Bh bohrium	108 Hs hassium	109 Mt meitnerium	110 Ds darmstadtium	111 Rg roentgenium	112 Cn copernicium
		5 B boron 11									
		6 C carbon 12									
		7 N nitrogen 14									
		8 O oxygen 16									
		9 F fluorine 19									
		10 Ne neon 20									
		13 Al aluminium 27									
		14 Si silicon 28									
		15 P phosphorus 31									
		16 S sulfur 32									
		17 Cl chlorine 35.5									
		18 Ar argon 40									
		31 Ga gallium 70									
		32 Ge germanium 73									
		33 As arsenic 75									
		34 Se selenium 79									
		35 Br bromine 80									
		50 Sn tin 119									
		51 Sb antimony 122									
		52 Te tellurium 128									
		53 I iodine 127									
		81 Tl thallium 204									
		82 Pb lead 207									
		83 Bi bismuth 209									
		84 Po polonium —									
		85 At astatine —									
		116 Lv livermorium —									
		117 Ts tennessine —									
		118 Og oganeson —									
		2 He helium 4									

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

lanthanoids

actinoids

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6 This question is about organic compounds.

(a) Organic compound **Q** has the following composition by mass.

C, 50.00%; H, 5.56%; O, 44.44%

Calculate the empirical formula of compound **Q**.

empirical formula = [3]

(b) Organic compound **R** has the empirical formula CHO and a relative molecular mass of 116.

Determine the molecular formula of compound **R**.

molecular formula = [1]

(c) Carboxylic acids react with alcohols to form esters.

(i) Name the other product that is formed when a carboxylic acid reacts with an alcohol.

..... [1]

(ii) Name the **type** of catalyst that is used when a carboxylic acid reacts with an alcohol.

..... [1]

(iii) Ester **S** has the structural formula $\text{CH}_3\text{CH}_2\text{COOCH}_2\text{CH}_2\text{CH}_2\text{CH}_3$.

Name the carboxylic acid and the alcohol which react together to form ester **S**.

carboxylic acid

alcohol

[2]





(d) Fig. 6.1 shows part of a polymer structure.

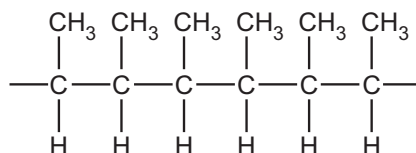


Fig. 6.1

- (i) Name the type of polymerisation that is used to produce this polymer.
..... [1]
- (ii) Suggest why it is **not** possible to write the molecular formula of this polymer.
..... [1]
- (iii) State the number of monomer units that are needed to make the part of the polymer structure shown in Fig. 6.1.
..... [1]
- (iv) Draw the displayed formula of the monomer used to make this polymer.

[2]





(e) Fig. 6.2 shows part of the general structure of an amino acid.

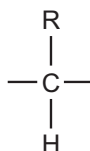


Fig. 6.2

(i) Complete Fig. 6.2 to show all the atoms and all the bonds in the two functional groups of the amino acid. [2]

(ii) Part of a natural polyamide structure is shown in Fig. 6.3.

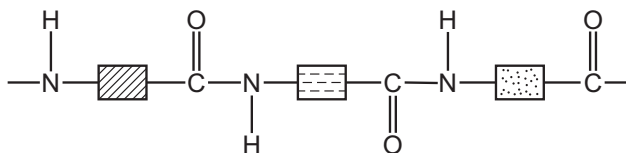


Fig. 6.3

On Fig. 6.3, draw a circle around **one** amide linkage. [1]

(iii) State the name given to natural polyamides made from amino acids. [1]

.....

[Total: 17]

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